

File No.: TOPBAND_Protocol_ModBus Revision:
V1.3

BMS ModBus Protocol

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Change History:

[illegible]

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1. CRC parity

CRC parity range is the check of all bytes before CRC field. It uses 16 bit CRC parity.

1.1 High significant byte CRC value

```
static unsigned char auchCRCHI[] = {  
    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,  
    0x00, 0xC1, 0x81,  
    0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81,  
    0x40, 0x01, 0xC0,  
    0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1,  
    0x81, 0x40, 0x01,  
    0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01,  
    0xC0, 0x80, 0x41,  
    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,  
    0x00, 0xC1, 0x81,  
    0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80,  
    0x41, 0x01, 0xC0,  
    0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,  
    0x80, 0x41, 0x01,  
    0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00,  
    0xC1, 0x81, 0x40,  
    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,  
    0x00, 0xC1, 0x81,  
    0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81,  
    0x40, 0x01, 0xC0,  
    0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,  
    0x80, 0x41, 0x01,  
    0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01,  
    0xC0, 0x80, 0x41,  
    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,  
    0x00, 0xC1, 0x81,  
    0x40  
};
```

1.2 Low significant byte CRC value

```
static char auchCRCLo[] = {
0x00, 0xC0, 0xC1, 0x01, 0xC3, 0x03, 0x02, 0xC2, 0xC6, 0x06, 0x07, 0xC7,
0x05, 0xC5, 0xC4,
0x04, 0xCC, 0x0C, 0x0D, 0xCD, 0x0F, 0xCF, 0xCE, 0x0E, 0x0A, 0xCA, 0xCB,
0x0B, 0xC9, 0x09,
0x08, 0xC8, 0xD8, 0x18, 0x19, 0xD9, 0x1B, 0xDB, 0xDA, 0x1A, 0x1E, 0xDE,
0xDF, 0x1F, 0xDD,
0x1D, 0x1C, 0xDC, 0x14, 0xD4, 0xD5, 0x15, 0xD7, 0x17, 0x16, 0xD6, 0xD2,
0x12, 0x13, 0xD3,
0x11, 0xD1, 0xD0, 0x10, 0xF0, 0x30, 0x31, 0xF1, 0x33, 0xF3, 0xF2, 0x32,
0x36, 0xF6, 0xF7,
0x37, 0xF5, 0x35, 0x34, 0xF4, 0x3C, 0xFC, 0xFD, 0x3D, 0xFF, 0x3F, 0x3E,
0xFE, 0xFA, 0x3A,
0x3B, 0xFB, 0x39, 0xF9, 0xF8, 0x38, 0x28, 0xE8, 0xE9, 0x29, 0xEB, 0x2B,
0x2A, 0xEA, 0xEE,
0x2E, 0x2F, 0xEF, 0x2D, 0xED, 0xEC, 0x2C, 0xE4, 0x24, 0x25, 0xE5, 0x27,
0xE7, 0xE6, 0x26,
0x22, 0xE2, 0xE3, 0x23, 0xE1, 0x21, 0x20, 0xE0, 0xA0, 0x60, 0x61, 0xA1,
0x63, 0xA3, 0xA2,
0x62, 0x66, 0xA6, 0xA7, 0x67, 0xA5, 0x65, 0x64, 0xA4, 0x6C, 0xAC, 0xAD,
0x6D, 0xAF, 0x6F,
0x6E, 0xAE, 0xAA, 0x6A, 0x6B, 0xAB, 0x69, 0xA9, 0xA8, 0x68, 0x78, 0xB8,
0xB9, 0x79, 0xBB,
0x7B, 0x7A, 0xBA, 0xBE, 0x7E, 0x7F, 0xBF, 0x7D, 0xBD, 0xBC, 0x7C, 0xB4,
0x74, 0x75, 0xB5,
0x77, 0xB7, 0xB6, 0x76, 0x72, 0xB2, 0xB3, 0x73, 0xB1, 0x71, 0x70, 0xB0,
0x50, 0x90, 0x91,
0x51, 0x93, 0x53, 0x52, 0x92, 0x96, 0x56, 0x57, 0x97, 0x55, 0x95, 0x94,
0x54, 0x9C, 0x5C,
0x5D, 0x9D, 0x5F, 0x9F, 0x9E, 0x5E, 0x5A, 0x9A, 0x9B, 0x5B, 0x99, 0x59,
0x58, 0x98, 0x88,
0x48, 0x49, 0x89, 0x4B, 0x8B, 0x8A, 0x4A, 0x4E, 0x8E, 0x8F, 0x4F, 0x8D,
0x4D, 0x4C, 0x8C,
0x44, 0x84, 0x85, 0x45, 0x87, 0x47, 0x46, 0x86, 0x82, 0x42, 0x43, 0x83,
0x41, 0x81, 0x80, 0x40
};
```

1.3 Program reference of realization

```
unsigned short CRC16 ( puchMsg, usDataLen ) /* The function returns the CRC as a
unsigned short type */ unsigned char *puchMsg ; /* message to calculate CRC upon */
unsigned short usDataLen ; /* quantity of bytes in message */
{
    unsigned char uchCRCHi = 0xFF ; /* high byte of CRC initialized */
    unsigned char uchCRCLo = 0xFF ; /* low byte of CRC initialized */
    unsigned uIndex ; /* will index into CRC lookup table */
    while (usDataLen--) /* pass through message buffer */
    {
        uIndex = uchCRCLo ^ *puchMsg++ ; /* calculate the CRC */
        uchCRCLo = uchCRCHi ^ auchCRCHi[uIndex] ;
        uchCRCHi = auchCRCLo[uIndex] ;
    }
    return (uchCRCHi<< 8 | uchCRCLo) ;
}
```

Code: 《MODBUS over Serial Line Specification and Implementation Guide V1.02》

1. Communication Parameters

BaudRate: **9600**

Parity bit: No

Data Bits: 8

Stop Bit: 1

communication address: BMS code switch value range 0~15, The BMS communication base address 38 , BMS communication address is sum of code switch value and communication base address.

Such as the code switch value is 1, so BMS communication address is 39.

2. Frame format of communication data

3.1 List of function code

Function code	submenu Function code	Meaning	Notes
0x03		Read command & product information	support to read single or Multi resister sequentially
0x11		Reading the Vendor Information	Model of product, software version, hardware version, serial number
0x12		Controls BMS sleep, wake, and reset	

Note:

1. Function code 0x03 means that sub node upload battery pack information collected to host node when sub node accept command 0x03 from host node;

2. Function code 0x11 means sub node load product information to host node when sub node accept command 0x11 from host node;

3. Function code 0x12: please check below:

Software--》BMS:

	Address	Command	Byte	Subcommand	CRC16 Check
	1Byte	1Byte	1Byte	1Byte	2Byte(LSB,MSB)
Sleep Command	addree	0x12	01	0xF0	CRC
Reset Command	addree	0x12	01	0xF1	CRC
Wake Command	addree	0x12	01	0xF2	CRC

BMS Response:

	Address	Command	Byte	Subcommand	CRC16 Check
	1Byte	1Byte	1Byte	1Byte	2Byte(LSB,MSB)
Sleep Command	addree	0x12	01	0xF0	CRC
Reset Command	addree	0x12	01	0xF1	CRC
Wake Command	addree	0x12	01	0xF2	CRC

Remark: Note: Send a control command, BMS executes the corresponding action, and returns the corresponding data frame;

3.2 Read the collected information frame format

Note 1: Type of command 0x03

Note 2: MSB means high significant byte, LSB means low significant byte.

Note 3: Each register store two bytes; Register data type is one byte of data, it is required to store in (LSB).

3.2.1 Host node sending frame format:

Item	0	1	2	3	4	5	6	7
Field definition	ADDR	CMD	MSB	LSB	MSB	LSB	LSB	MSB
Explanation	Controller address	Type of command	beginning register address		Resister number n		CRC parity	

3.2.2 Sub node normal response of frame format:

Item	0	1	2	3	4	5	6	...	L+1	L+2	L+3	L+4
Field definition	ADDR	CMD	Length	MS B	LS B	MS B	LS B	...	MS B	LS B	LS B	MS B
Explanation	Controller address	Type of command	Bytes transferred $L=n*2$	First register's value		Second register's value		...	Last register's value		CRC parity	

3.2.3 Abnormal response format from sub node:

Item	0	1	2	3	4
Field definition	ADDR	CMD+128	ErrCode	LSB	MSB
Explanation	Controller address	Type of command +128	Error Code = 1	CRC parity	

Note: CRC parity range is the check of all bytes before CRC field.

3.3 Read product information frame format

Note 1: Type of command 0x11

Note 2: MSB means high significant byte, LSB means low significant byte.

Note 3: Each register store two bytes; Register data type is one byte of data, it is required to store in (LSB).

3.3.1 Host node sending frame format:

Item	0	1	2	3	4	5	6	7
Field definition	ADDR	CMD	MSB	LSB	MSB	LSB	LSB	MSB
Explanation	Controller address	Type of command	beginning register address		Resister number n		CRC parity	

3.3.2 Normal response of frame format from sub node:

Item	0	1	2	3	4	5	6	...	L+1	L+2	L+3	L+4
Field definition	ADDR	CMD	Length	BYTE1 ... BYTE _n							LSB	MSB
Explanation	Controller address	Type of command	Bytes transferred L=n	Model of product+'*' + software version+'*' + hardware version+'*' + serial number+'*'							CRC parity	

Note: Each content should be separated by '*' of model of product, software version, hardware version, serial number

3.3.3 Abnormal response of format from sub node:

Item	0	1	2	3	4
Field definition	ADDR	CMD+128	Err Code	LSB	MSB
Explanation	Controller address	Type of command +128	Error Code = 1	CRC parity	

Note: CRC parity range is the check of all bytes before CRC field.

4. Data information

4.1 Data acquisition

Note 1: If the reading data is invalid value, reported 0xFFFF

Note 2: Each register stores two bytes

RegId	Data	Unit	Bytes	Scale Factor	Offset	PDU#Value	Min data value	Max data value
0x1000	Voltage of battery pack	V	2	0.01	0	5343#53.43	0	64
0x1001	current of battery pack	A	2	0.1	-10000	9505# -49.5	-100	100
0x1002	Remaining capacity	Ah	2	0.1	0	560#56	0	200
0x1003	Average temperature of cell	°C	2	0.1	-400	300# -10	-40	120
0x1004	Environment temperature	°C	2	0.1	-400	300# -10	-40	120
0x1005	Warning Flag	#	2	0.1	0	See description	0x0000	0xffff
0x1006	Protection Flag	#	2	0.1	0	See description	0x0000	0xffff

0x1007	Fault/Status	#	2	0.1	0	See description	0x0000	0xffff
0x1008	SOC	#	2	0.01 %	0	12.34#1234	0	100
0x1009	Average charge voltage	V	2	0.01	0			
0x100A	Float charge voltage	V	2	0.01	0			
0x100B	Maximum charging current	A	2	0.1	-10000			
0x100C	Maximum discharging current	A	2	0.1	-10000			
0x100D	SOH		2	0.01%	0			
0x100E	Full Charge Capacity	Ah	2	0.1	0			
0x100F	Current PACK address	/	2	1	0			
0x1010	Cycle Times	/	2	1	0			
0x1011~0x1020	Voltage from cell 1to cell 16	mV	32	1	0			
0x1021~0x1024	Cell temperature 1 to 4	°C	8	0.1	-400			
0x1025	Balance Temperature	°C	2	0.1	-400			
0x1026	MOS Temperature	°C	2	0.1	-400			
0x1027	Battery Pack overvoltage alarm value	V	2	0.01	0			
0x1028	Battery Pack undervoltage alarm value	V	2	0.01	0			
0x1029	Maximum Cell Voltage	mV	2	1	0			
0x102A	Minimum Cell Voltage	mV	2	1	0			

4.2 Information of product

	number of bytes	Example	Note
Model of product	10 (max number of bytes)	LD4850 - X	ASCII transmission

software version	2 (fixed)	0x0A0A , version : V10.10	16 hexadecimal code, version (1 byte) +revision (1 byte)
hardware version	5 (fixed)	0x010a0b0200 version : V01.10.11.02.00	16 hexadecimal code
serial number	20 (fixed)	20151111011800400026	ASCII transmission

5. Instruction

5.1 Warning Flag

Byte0	Bit0	1:battery cell overvoltage alarm 0:not occurring	Not suggest to display
	Bit1	1: battery cell low voltage alarm 0: not occurring	Not suggest to display
	Bit2	1: battery pack overvoltage alarm 0: not occurring	Not suggest to display
	Bit3	1:battery pack low voltage alarm 0: not occurring	Not suggest to display
	Bit4	1: charging over current alarm 0: not occurring	Not suggest to display
	Bit5	1:discharging over current alarm 0: not occurring	Not suggest to display
	Bit6	1: battery high temperature alarm 0: not occurring	Not suggest to display
	Bit7	1: battery low temperature alarm 0: not occurring	Not suggest to display
Byte1	Bit0	1:environment high temperature alarm 0: not occurring	Not suggest to display
	Bit1	1: environment low temperature alarm 0: not occurring	Not suggest to display
	Bit2	1:MOS high temperature alarm 0: not occurring	Not suggest to display
	Bit3	1:low capacity alarm 0: not occurring	Not suggest to display
	Bit4	reserve	
	Bit5	reserve	
	Bit6	reserve	
	Bit7	reserve	

5.2 Protection Flag

Byte0	Bit0	1: battery cell over voltage protection 0: not occurring	Not suggest to display
	Bit1	1: battery cell low voltage protection 0: not occurring	Not suggest to display
	Bit2	1: battery pack over voltage protection 0: not occurring	Not suggest to display
	Bit3	1: battery pack low voltage protection 0: not occurring	Not suggest to display
	Bit4	1: short circuit protection 0: not occurring	
	Bit5	1: over current protection 0: not occurring	
	Bit6	1: charging high temperature protection 0: not occurring	
	Bit7	1: charging low temperature protection 0: not occurring	
Byte1	Bit0	1: discharging high temperature protection 0: not occurring	
	Bit1	1: discharging low temperature protection 0: not occurring	
	Bit2	reserve	
	Bit3	reserve	
	Bit4	reserve	
	Bit5	reserve	
	Bit6	reserve	
	Bit7	reserve	

5.3 Fault Flag/Status Flag

Byte0 (Fault)	Bit0	1: front end sampling communication fault 0: not occurring	
	Bit1	1: temperature sensor break 0: not occurring	
	Bit2	reserve	
	Bit3	reserve	
	Bit4	reserve	
	Bit5	reserve	
	Bit6	reserve	
	Bit7	reserve	
Byte1 (Status)	Bit0	1: state of charge 0: not occurring	
	Bit1	1: state of discharge 0: not occurring	
	Bit2	1: charge MOS break over 0: charge MOS break	
	Bit3	1: discharge MOS break over 0: discharge MOS break	
	Bit4	1: charge limit current open 0: charge limit current close	
	Bit5	1: The heating film open 0: The heating film close	
	Bit6	reserve	
	Bit7	reserve	